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CEMTNet: a cognitive emotion modulated network for multimodal depression detection

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ABSTRACT

Early depression detection is a critical task in public health, making automatic depression identification increasingly important. Existing multimodal methods face challenges of cross-modal emotional inconsistency, subtle emotional cue extraction, and dynamic emotional modeling. To address these issues, we propose Cognitive-Emotion-Modulated Trend Network (CEMTNet) for robust multimodal depression recognition from audio-text data. CEMTNet integrates three modules: Cognitive Consistency Inference Mechanism (CCIM) dynamically adjusts fusion weights based on estimated emotional consistency; Emotion-Modulated Structured Attention (EMSA) enhances fine-grained emotional cue extraction using emotion-aware multi-head attention; and Contrastive Emotion Trend Modeling (CE-TM) integrates contrastive learning with temporal trend modeling. The experimental results on the DAIC-WOZ and EATD-Corpus datasets demonstrate that CEMTNet consistently outperforms existing state-of-the-art methods, achieving an accuracy of 0.92, an F1-score of 0.92, and a recall of 0.93. The ablation studies and case analyses further confirm the effectiveness and robustness of the proposed modules. The source code can be found at <https://github.com/ghy-yhg/CEMTNet>.

1. Introduction

Depression is a prevalent and severely debilitating mental health disorder that significantly impairs emotional regulation, cognitive functioning, and social adaptability (Malhi & Mann, 2018; Marcus et al., 2012). According to the World Health Organization (WHO) (World Health Organization, 2008), over 300 million individuals worldwide suffer from depression, making it a leading cause of disability and disease burden. However, due to the heterogeneity and subtlety of depressive symptoms, many cases remain undiagnosed or untreated, with severe conditions potentially escalating to self-harm or suicide (Visted et al., 2018). In current psychiatric practice, diagnosis typically relies on clinical assessment combined with standardized self-report instruments, such as the Beck Depression Inventory (BDI) and the Patient Health Questionnaire (PHQ) (BECK et al., 1961; Kroenke et al., 2001). Nonetheless, these approaches are susceptible to reporting biases and constrained by an increasing imbalance between clinical demand and available resources, adversely impacting diagnostic efficiency and reliability. Therefore, developing automated, objective, and scalable depression detection technologies has emerged as a viable solution to

overcome these limitations (Fang et al., 2023a; Kim et al., 2023; Kumari et al., 2025; Mao et al., 2023). Existing automated approaches generally fall into two categories: physiological and non-physiological signal-based methods. Physiological signals, including heart rate (Nepal et al., 2024), electrodermal activity (Hickey et al., 2021), and electroencephalography (Shen et al., 2023), can capture internal biological changes but require specialized sensors, complicating their deployment. In contrast, non-physiological signals such as facial images (Ren et al., 2025), text (Chiong et al., 2021; Phiri et al., 2025), and audio (Almaghrabi et al., 2023; Miao et al., 2022) reflect emotional states and are easier to continuously monitor. Audio and text modalities, favored due to ease of collection and fewer privacy concerns (van Heerden et al., 2024; Mundt et al., 2012), have become focal points in current depression detection research. As illustrated in Fig. 1, depression detection and assessment can be effectively achieved through multimodal integration of textual and audio data.

In recent years, both textual and audio modalities have demonstrated effectiveness in scenarios such as social media analyses and clinical interviews (Wang et al., 2019). However, depression inherently possesses multidimensional and cross-modal characteristics;

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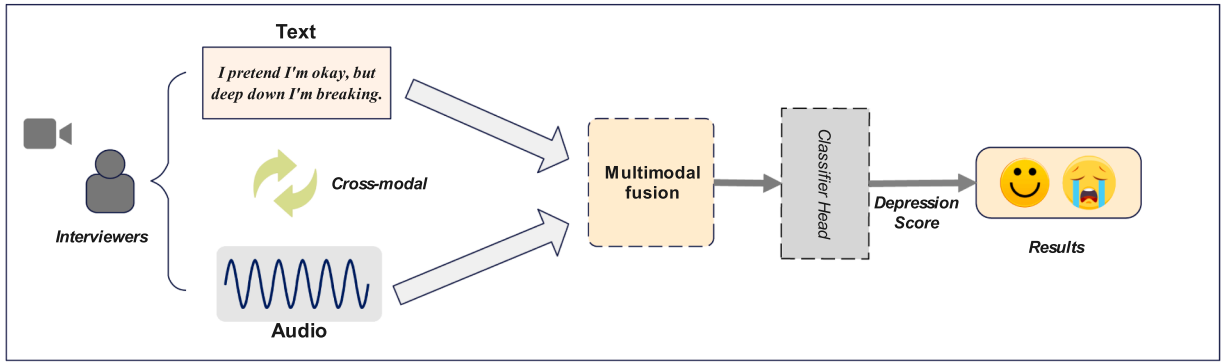


Fig. 1. The illustration of multimodal depression detection integrating audio with text modalities.

single-modality methods often overlook critical integrated cues (He et al., 2022; Liu et al., 2025b; Tian et al., 2024; Wang et al., 2019). Ye et al. designed the SESE experiment to capture subtle differences between audio and text arising from rapid emotional transitions but did not resolve semantic conflicts between modalities (Ye et al., 2021). Zhang et al. proposed the AVTF-TBN model, employing attention mechanisms and residual connections to dynamically fuse multimodal features and preserve differences in feature importance. However, this approach lacks explicit semantic alignment, causing subtle emotional signals to be overlooked (Zhang et al., 2024c). Additionally, Ding et al. introduced IntervoxNet, utilizing Transformer-based self-attention to capture long-range dependencies between audio and text modalities. Their Long Short-Term Attention Fusion Network (LSTAFN) enabled dynamic cross-modal alignment, but primarily focused on static interview scenarios, failing to sufficiently model nonlinear emotional dynamics over time (Ding et al., 2024).

To address these challenges, we propose a Cognitive-Emotion Modulated Trend Network (CEMTNet) for multimodal depression detection, consisting of three integrated modules. First, the Cognitive Consistency Inference Mechanism (CCIM) receives projected textual and audio modality features as input, dynamically quantifying cross-modal emotional consistency through a consistency scoring network. Using these scores, a gating weight generation module adaptively computes fusion weights, suppressing modality conflicts and enhancing consistency to produce stable, coordinated multimodal emotional representations. These fused features are then fed into the Emotion-Modulated Structured Attention (EMSA) module, which integrates learnable emotion-guided vectors within a multi-head attention framework. EMSA further incorporates positional biases and attention masks to encode temporal information and filter noise, accurately extracting subtle and dynamic depressive emotional cues. After attention-weighted context aggregation, EMSA outputs an emotion-aware sequence feature for subsequent dynamic modeling. Lastly, the Contrastive Emotion Trend Modeling (CE-TM) module processes these sequence features using a Long Short-Term Memory (LSTM) network, capturing emotional dynamics across time steps and modeling emotional transitions and accumulations. Experimental evaluations on two benchmark datasets, DAIC-WOZ (Gratch et al., 2014) and EATD-Corpus (Shen et al., 2022), demonstrate that CEMTNet consistently outperforms competitive baselines, achieving a maximum accuracy of 93%, an F1-score of 0.93, and a recall of 0.92. Ablation studies further confirm the individual and combined contributions of the proposed modules.

In summary, the major contributions of this work are as follows:

- This study introduces the CCIM, which dynamically regulates cross-modal emotion coherence. Unlike previous methods that overlook semantic dissonance between modalities, CCIM explicitly aligns sentiment orientation across text and audio, filtering out conflicting affective signals and enhancing emotional consistency perception.

- We propose an EMSA module, which introduces position bias and attention mask based on traditional multi-head attention, and combines the structured attention mode guided by emotion prior to amplify weak and ambiguous emotions. The model is able to detect hesitation, flat tone or emotionally ambiguous language.
- We designed a CE-TM module, which inputs the emotion time series enhancement feature sequence into the LSTM and deeply integrates the contrastive learning idea with emotion time series trend modeling.
- Extensive experiments were conducted on two publicly available clinical interview datasets to evaluate the effectiveness of the proposed approach. The results demonstrate that our method outperforms existing state-of-the-art (SOTA) techniques in multimodal depression detection, highlighting its potential for facilitating early diagnosis of depression in clinical applications.

2. Related work

We review related studies along two central challenges that directly motivate our proposed framework: (1) multimodal fusion with cross-modal consistency, and (2) modeling weak emotional cues with temporal dynamics.

2.1. Multimodal fusion and cross-modal consistency challenges

Multimodal depression detection aims to integrate heterogeneous modalities-such as text, audio, and visual signals-to capture the dynamic representation of complex depressive states. Although significant progress has been made in existing fusion frameworks, they remain limited by core challenges related to cross-modal semantic misalignment and emotional inconsistency.

Recent studies have explored dynamic fusion strategies, such as the Task-Adaptive Cross-modal Fusion Network (TACFN) (Liu et al., 2023), MLP-Mixer-based audio feature optimization models (Lim et al., 2025), and dual attention networks (HMATN) that reinforce audio-visual associations (Moorthy & Moon, 2025). Hybrid CNN-BiLSTM architectures (Zhang et al., 2023) and joint text-audio modeling approaches (Morales et al., 2018) have further enhanced spatiotemporal feature representations, while methods such as CANARMRF (Wei et al., 2024) and deep multimodal feature extraction frameworks (Khowaja et al., 2025) have improved fusion accuracy. However, most of these approaches primarily rely on early or late fusion paradigms, inherently limiting their ability to explore deep semantic alignment across heterogeneous modalities. Moreover, due to limited discriminative capacity, low-intensity emotional cues (e.g., negative textual content delivered with neutral prosody) often fail to receive sufficient attention weight under conventional attention mechanisms, leading to the neglect of clinically critical indicators for depression detection.

Cross-modal emotional inconsistency represents another core challenge, wherein conflicting emotional tendencies exist across different

modalities (Sadeghi et al., 2024; Wang et al., 2024). While certain models have enhanced unimodal representational capacity through improved feature extraction techniques (Huang et al., 2024; Wang et al., 2023; Zhang et al., 2024a), most fusion methods lack explicit emotional consistency alignment mechanisms. For instance, approaches such as RLKT-MDD (Yang et al., 2024) suffer from imbalanced modality quality, while others (Wu et al., 2024; Zhang et al., 2024b) often yield unstable fusion results under noisy or unbalanced input conditions due to the absence of adaptive modulation mechanisms.

To address these limitations, this study proposes the CCIM, which dynamically evaluates emotional consistency across modalities and suppresses conflicting signals during the fusion process, thereby enhancing cross-modal semantic coordination and improving model robustness.

2.2. Weak emotional cues and temporal dynamic modeling

In clinical settings, depressive emotions often manifest as low-intensity and ambiguous emotional cues, such as flattened intonation, slowed speech rate, and vague linguistic expressions. These subtle signals pose significant challenges for existing models in terms of sensitivity and feature focus. Moreover, depressive symptoms exhibit nonlinear temporal evolution, necessitating dynamic sequential modeling approaches beyond static classification frameworks.

To address the challenge of weak cue representation, several studies have employed hierarchical structures to improve feature granularity. Techniques such as emotional distribution modeling (Inoue et al., 2024), hierarchical attention mechanisms (Zhang et al., 2025), and multi-source attention fusion methods like MDHAN (Zogan et al., 2022) have enhanced multi-level feature representation capabilities but still struggle to effectively capture non-prominent cues. Other methods, including cross-modal mutual information maximization (Han et al., 2021), redundant feature compression (Fang et al., 2023b), and self-supervised learning approaches (Su et al., 2025; Yu et al., 2021), often overlook the capture of fine-grained emotional signals. While micro-expression analysis (Li et al., 2024b) and acoustic feature engineering (Yang et al., 2023) focus on local signals, they lack global cross-modal regulation mechanisms, resulting in the dilution of key cues during the fusion process.

In terms of temporal dynamic modeling, sequential models such as Long Short-Term Memory (LSTM), Bidirectional Gated Recurrent Unit (BiGRU), Temporal Convolutional Networks (TCN), and Transformer architectures have been widely employed to capture emotional dynamics (Alam et al., 2025; Huang et al., 2021; Raut et al., 2024; Salas-Caceres et al., 2024; Xing et al., 2024). However, most approaches focus exclusively on intramodal temporal dependencies and lack explicit modeling of cross-modal temporal alignment and nonlinear emotional transitions. Although recent advances such as attention-based methods (Liu et al., 2025a; Woo et al., 2025), multi-resolution fusion models (Raut et al., 2024), and hybrid dynamic-static networks (Liu et al., 2025a) have introduced improvements, they have yet to resolve the challenge of modeling asynchronous cross-modal trends. Shared space learning and physiological modality integration methods (Li et al., 2024a, 2022; Ma et al., 2025; Zhang et al., 2021) offer partial optimization but still fail to explicitly capture emotional trend inflection points and nonlinear evolutionary processes.

To overcome these limitations, this study introduces two core components: the EMSA module, which incorporates emotion priors to guide attention allocation and enhance the capture of weak depressive cues; and the CE-TM module, which integrates cross-timestep contrastive learning with state consistency constraints to explicitly model the nonlinear emotional evolution process involved in depression detection. Table 1 provides a comparative summary of these techniques, detailing their primary advantages and limitations. This comparison underscores the ongoing challenges in cross-modal alignment, weak cue extraction, and temporal modeling that motivate the design of CEMTNet.

3. Proposed method

3.1. Overall framework

To address the challenges of cross-modal emotional misalignment, weak emotional cues, and temporal dynamics in depression detection, we propose a novel framework CEMTNet. The architecture of CEMTNet and its internal components are illustrated in Fig. 2. CEMTNet consists of three synergistically integrated core modules: the CCIM, the EMSA, and the CE-TM.

Specifically, CCIM focuses on regulating modality-level emotional consistency to address semantic conflicts between audio and text inputs, while EMSA targets fine-grained emotional perception within each modality, enhancing the detection of subtle and weak emotional cues. CE-TM is designed to model the temporal evolution of emotional states through a contrastive learning mechanism, thereby improving the capture and discrimination of depression-related temporal patterns. This clear functional division enables the joint optimization of cross-modal coordination and intra-modal perception, ultimately enhancing the robustness of depression recognition. Furthermore, to prevent the loss of fine-grained emotional data during module integration, CEMTNet incorporates explicit feature preservation mechanisms. While pooling and attention operations are used to extract high-level semantics, critical temporal details are strictly retained by concatenating local frame-level features with global embeddings and utilizing residual connections in the final temporal modeling stage.

3.2. Cognitive consistency inference mechanism

The core innovation of the CCIM lies in integrating a trainable consistency scoring network with an adaptive gating fusion module to dynamically model and regulate cross-modal emotional consistency. In contrast to mechanisms that emphasize intra-modal perception, CCIM is dedicated to resolving emotional and semantic conflicts across modalities, ensuring affective coherence in the fusion process. This mechanism establishes a stable multimodal representation foundation that facilitates subsequent fine-grained modeling in downstream modules. Initially, the emotional alignment between text and audio modalities is quantified to provide decision guidance for subsequent dynamic fusion. The raw modality-specific features are first projected into high-level semantic representations through a two-layer multilayer perceptron (MLP), and then compressed into compact unimodal emotion vectors via global average pooling. These vectors are concatenated and fed into a trainable similarity estimator, which applies nonlinear mapping to produce a normalized consistency score representing the sample-level cross-modal emotional alignment: higher scores indicate strong sentiment alignment, while lower scores suggest potential semantic conflicts. Unlike traditional similarity-based or static fusion approaches, the trainable similarity estimator learns complex cross-modal consistency patterns through end-to-end optimization. Based on the consistency score, the gating weight generation module concatenates the score with the unimodal emotion vectors and passes them through a gating network to compute dynamic fusion weights. This adaptive weighting mechanism suppresses contributions from less reliable modalities when conflicts are detected, while emphasizing semantically aligned signals. Finally, the consistency-aware fusion operation integrates the multimodal emotional representations according to the learned weights, generating harmonized features for downstream modules. The complete computational procedure of the module can be expressed as:

$$H_{\text{text}} = \text{MLP}(X_{\text{text}}), \quad H_{\text{audio}} = \text{MLP}(X_{\text{audio}}) \quad (1)$$

$$T = \text{GAP}(H_{\text{text}}), \quad A = \text{GAP}(H_{\text{audio}}) \quad (2)$$

$$S = \text{SimNet}(\text{Concat}(T, A)) \quad (3)$$

$$G = \text{GateGen}(\text{Concat}(T, A, S)) \quad (4)$$

Table 1

Comparison of representative methodologies for multimodal depression detection.

Methodology	Refs.	Advantages	Limitations
Multimodal Fusion	(Khowaja et al., 2025; Lim et al., 2025; Liu et al., 2023; Moorthy & Moon, 2025; Morales et al., 2018; Wei et al., 2024; Zhang et al., 2023)	Enhances spatiotemporal features to enable dynamic cross-modal interactions.	Lacks deep semantic alignment and yields unstable results under noisy inputs.
Unimodal & Local Features	(Huang et al., 2024; Li et al., 2024b; Wang et al., 2023; Yang et al., 2023; Zhang et al., 2024a)	Improves unimodal capacity by effectively capturing local acoustic and visual signals.	Lacks global cross-modal regulation which causes the dilution of key cues during fusion.
Cross-Modal Consistency	(Sadeghi et al., 2024; Wang et al., 2024; Yang et al., 2024)	Mitigates conflicting emotional tendencies across different data modalities.	Suffers from modality imbalance and lacks explicit emotional consistency alignment.
Attention & Hierarchical	(Inoue et al., 2024; Liu et al., 2025a; Woo et al., 2025; Zhang et al., 2025; Zogan et al., 2022)	Improves feature granularity through multi-source attention fusion architectures.	Overlooks non-prominent, low-intensity, and ambiguous emotional cues.
Temporal Modeling	(Alam et al., 2025; Huang et al., 2021; Raut et al., 2024; Salas-Cáceres et al., 2024; Xing et al., 2024)	Captures intramodal emotion dynamics and long-range sequence dependencies.	Fails to explicitly model asynchronous cross-modal temporal alignment.
Shared Space	(Fang et al., 2023b; Han et al., 2021; Li et al., 2024a, 2022; Ma et al., 2025; Su et al., 2025; Yu et al., 2021; Zhang et al., 2021)	Maximizes cross-modal mutual information while compressing redundant features.	Overlooks fine-grained signals and nonlinear emotional trend inflection points.

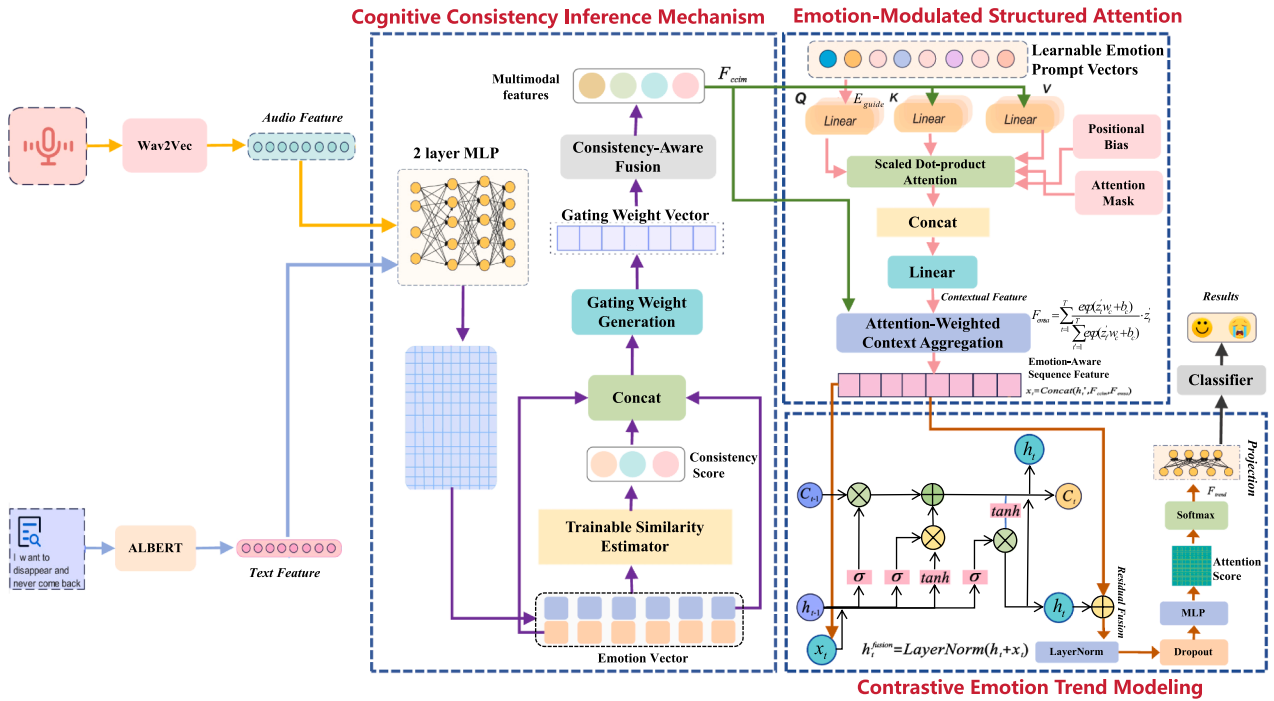


Fig. 2. Illustrates the proposed CEMENTNet, in which text features are extracted using A Lite BERT (ALBERT), while acoustic features are encoded via wav2vec 2.0. These modality-specific features are subsequently refined through three core modules: the CCIM, which dynamically aligns cross-modal emotional tendencies; the EMSA module, which leverages structured emotional priors to enhance weak and ambiguous emotional cues; and the CE-TM module, which captures the temporal evolution of emotional states across different utterances. The final fused representation is then fed into a classifier for depression prediction.

$$F_{ccim} = \text{Fusion}([T, A], G) \quad (5)$$

Let X_{text} and X_{audio} denote the original textual and acoustic input features, respectively. The operator $\text{MLP}(\cdot)$ represents a two-layer multi-layer perceptron, and $\text{GAP}(\cdot)$ denotes the global average pooling operation. The resulting pooled vectors T and A correspond to the aggregated textual and acoustic emotional embeddings. The consistency scoring network, denoted as SimNet, outputs the consistency score S via supervised learning. The gating weight generator is represented as GateGen,

which produces adaptive fusion weights. The consistency-aware fusion module is denoted as Fusion. Finally, the output F_{ccim} represents the adaptively fused multimodal emotional representation.

3.3. Emotion-modulated structured attention

The core innovation of the EMSA module lies in its proposed emotion-guided structured attention mechanism. The primary objective

of this module is to model weak or ambiguous emotional signals within individual modalities. Built upon the conventional multi-head attention, EMSA introduces positional bias and attention mask mechanisms to significantly enhance the model's capability in capturing fine-grained emotional cues. In addition, an attention-weighted context aggregation strategy is developed to dynamically integrate global emotional guidance with contextual information, enabling multi-granularity emotional modeling and enhancing emotion cue expressiveness.

As illustrated in Fig. 2, the module first incorporates learnable emotion prompt vectors to guide attention, facilitating the identification of emotion-sensitive regions in the high-dimensional feature space. These guided vectors are then combined with the fused features and linearly projected to generate query (Q), key (K), and value (V) representations. During attention computation, positional bias is introduced into the scaled dot-product attention to encode temporal positional information, enhancing the model's ability to capture emotional transitions over time. Meanwhile, the attention mask suppresses noise and irrelevant fragments during attention weight allocation. The multi-head attention outputs are concatenated and projected through a linear layer to obtain contextual features, which are subsequently aggregated by the attention-weighted context aggregation module to produce the final emotion-aware sequence feature. In contrast to the modality alignment-oriented CCIM, EMSA focuses on fine-grained emotional modeling within individual modalities. It is particularly effective in capturing depressive manifestations such as hesitation, flat intonation, or ambiguous emotional expressions, thereby enhancing the precision and sensitivity of emotion recognition. The corresponding mathematical formulations are described as follows:

First, the learnable emotion prompt vectors are initialized as:

$$E_{\text{guide}} = \{e_1, e_2, \dots, e_N\}, \quad e_i \in \mathbb{R}^d \quad (6)$$

where E_{guide} denotes the set of emotion prompts, N is the number of prompts, and d is the feature dimension.

Next, the emotion prompt vectors and the fused features are projected into Q , K , and V spaces, and the structured attention output is computed as:

$$Q = W_Q E_{\text{guide}}, \quad K = W_K F_{\text{ccim}}, \quad V = W_V F_{\text{ccim}} \quad (7)$$

$$H = \text{Softmax}\left(\frac{QK^T + B_{\text{pos}} + M_{\text{mask}}}{\sqrt{d}}\right)V \quad (8)$$

$$H' = \text{Linear}(\text{Concat}(H_1, H_2, \dots, H_N)) \quad (9)$$

where Q , K , and V represent the query, key, and value matrices, respectively; W_Q , W_K , and W_V are projection weight matrices; B_{pos} denotes the positional bias; M_{mask} denotes the attention mask; A represents the attention weights; H is the attention output; $\text{Linear}(\cdot)$ denotes the linear transformation; $\text{Concat}(\cdot)$ denotes concatenation operation; and H' represents the multi-prompt contextual feature after aggregation.

Subsequently, for each time step, the contextual feature h'_t is concatenated with the global cognitive consistency representation F_{ccim} to form the frame-wise aggregated representation z_t , which is computed as:

$$z_t = \text{Concat}(h'_t, F_{\text{ccim}}) \quad (10)$$

where h'_t denotes the contextual feature at time step t , which is extracted from the multi-prompt attention output H' after linear transformation, and F_{ccim} is the global cognitive consistency embedding.

The aggregated feature z_t is then projected through a linear transformation to obtain z'_t , which is formulated as:

$$z'_t = W_{\text{agg}} z_t + b_{\text{agg}} \quad (11)$$

where z_t is the frame-wise concatenated feature at time step t , formed by integrating the contextual feature h'_t and the global cognitive consistency representation F_{ccim} . The vector z'_t is obtained by linearly projecting z_t with learnable parameters W_{agg} and b_{agg} .

To aggregate the temporal contextual features into a unified emotional representation, we employ a soft attention mechanism based on the projected features z'_t , computed as:

$$F_{\text{emsa}} = \sum_{t=1}^T \frac{\exp(z'_t w_c + b_c)}{\sum_{t'=1}^T \exp(z'_{t'} w_c + b_c)} \cdot z'_t \quad (12)$$

where w_c and b_c are learnable pooling parameters that determine the attention weights, and F_{emsa} denotes the emotion-aware global representation, aggregated as a weighted sum over all time steps.

3.4. Contrastive emotion trend modeling

To model the temporal evolution of depressive emotional states, we propose the CE-TM module, which processes a sequence of emotion-aware contextual features. Each input x_t is constructed by concatenating the frame-level contextual vector h'_t , the global cognitive embedding F_{ccim} , and the global emotion-aware feature F_{emsa} aggregated in the EMSA module:

$$x_t = \text{Concat}(h'_t, F_{\text{ccim}}, F_{\text{emsa}}) \quad (13)$$

where h'_t is the contextual feature at time step t from EMSA output, F_{ccim} is the global cognitive-consistency representation, and F_{emsa} is the global emotion-aware representation.

The sequence $\{x_t\}_{t=1}^T$ is fed into a Long Short-Term Memory (LSTM) network to capture long-range dependencies. At each time step t , the hidden state h_t and memory cell C_t are updated, and a residual connection is applied, followed by Layer Normalization:

$$(h_t, C_t) = \text{LSTM}(x_t, h_{t-1}, C_{t-1}; \theta_{\text{Lstm}}), \quad h_t^{\text{fusion}} = \text{LayerNorm}(h_t + x_t) \quad (14)$$

Here, h_t, C_t represent the LSTM hidden and memory states, and h_t^{fusion} denotes the residual-enhanced LSTM output.

To highlight salient temporal segments, a soft attention mechanism computes weighted temporal fusion as follows:

$$F_{\text{trend}} = \sum_{t=1}^T \frac{\exp(\text{MLP}(h_t^{\text{fusion}})w_s + b_s)}{\sum_{t'=1}^T \exp(\text{MLP}(h_{t'}^{\text{fusion}})w_s + b_s)} \cdot h_t^{\text{fusion}} \quad (15)$$

where w_s, b_s are the learnable projection weights and bias, and F_{trend} is the aggregated emotional trend feature.

The resulting emotional trend representation F_{trend} is passed through dropout, projection, and softmax for final classification:

$$\hat{y} = \text{Softmax}(\text{MLP}(\text{Dropout}(F_{\text{trend}}))) \quad (16)$$

where $\hat{y}$ denotes the predicted depression class probability.

To improve inter-class separability, a contrastive loss is introduced:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda \sum_{(i,j)} \mathbb{I}(y_i \neq y_j) \cdot \max\left(0, m - \|F_{\text{trend}}^{(i)} - F_{\text{trend}}^{(j)}\|_2\right) \quad (17)$$

Here, $\mathcal{L}_{\text{cls}}$ is the cross-entropy classification loss, and the second term represents the inter-class contrastive loss with margin m and balance weight λ .

4. Experiments

4.1. Dataset

Experiments were conducted on two depression datasets, namely DAIC-WOZ and EATD-Corpus. The class distribution of depressed and non-depressed samples in each dataset is presented in Table 2.

4.1.1. DAIC-WOZ dataset

The DAIC-WOZ dataset was developed by the Gratch team (Gratch et al., 2014) as a multimodal research resource to support the clinical identification of anxiety, depression, and post-traumatic stress disorder

Table 2
Category distribution of depression datasets.

Dataset	Language	Participants	Nondepressed	Depressed
DAIC-WOZ (Gratch et al., 2014)	English	189	133	56
EATD-Corpus (Shen et al., 2022)	Chinese	162	132	30

Table 3
DAIC-WOZ dataset partition statistics.

Clinical Group	Training Set	Validation Set	Test Set
Healthy Subjects	77	23	33
Depressed Subjects	30	12	14
Total Samples	107	35	47

Table 4
EATD-Corpus dataset partition statistics.

Clinical Group	Training Set	Test Set
Healthy Subjects	118	14
Depressed Subjects	21	9
Total Samples	139	23

(PTSD). The dataset was collected through interactive interviews between participants and the virtual interviewer Ellie, during which diagnostically valuable behavioral and physiological features were simultaneously recorded. During the experiment, the audio modality recorded participants' prosodic features and speech fluency, while the text modality contained verbatim transcripts of the dialogues to analyze participants' language styles and semantic features. The experimental design emphasized the construction of natural dialogue scenarios to effectively extract depression-related indicators in real-world contexts.

The dataset includes 189 participants, who were categorized into a healthy control group (133 individuals) and a depression group (56 individuals) based on clinical diagnostic criteria. The data were further divided into training, validation, and test sets (see Table 3). A dual-channel recording system was used to independently record the interactions between each participant and Ellie, with temporal annotations applied to accurately segment the human-computer dialogue fragments. In the data preprocessing stage, the original discrete files were integrated to generate individual audio and corresponding text files for each participant, in which the text data were automatically annotated to distinguish between human and virtual agent utterances.

4.1.2. EATD-Corpus dataset

The EATD-Corpus is a Chinese depression dataset collected from 162 student volunteers at a university in China (Shen et al., 2022). The dataset includes participants' audio responses and corresponding text transcripts obtained from virtual interviews. For each participant, both audio and text files are provided, and the data are categorized into three classes: negative, neutral, and positive. The sample sizes are presented in Table 4.

In this study, to facilitate comparative experiments, features were extracted separately from the negative, neutral, and positive files and then combined to create a single audio feature and a single text feature for each participant, allowing the model to be trained and learned more effectively.

4.2. Evaluation metrics

In this study, commonly used evaluation metrics including Accuracy, Precision, Recall, and F1-score were adopted. Accuracy reflects the overall proportion of correct predictions made by the model; Precision indicates the proportion of truly depressed samples among those predicted as depressed; Recall measures the model's ability to correctly identify actual depressed individuals; and F1-score, as the harmonic mean of Precision and Recall, emphasizes a comprehensive evaluation of the model's

performance in recognizing the minority class. The mathematical definitions of these metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (19)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (20)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (21)$$

Here, TP denotes the number of samples correctly predicted as depressed; TN denotes the number of samples correctly predicted as non-depressed; FP denotes the number of samples incorrectly predicted as depressed but actually non-depressed; and FN denotes the number of samples incorrectly predicted as non-depressed but actually depressed.

4.3. Preprocessing and parameter settings

In the audio feature modeling part, the wav2vec 2.0 pretrained model was used to extract frame-level hidden features that capture acoustic properties, prosodic variations, and emotional information. For textual semantic modeling, the ALBERT model was employed to obtain high-level semantic representations with contextual awareness.

4.3.1. wav2vec 2.0

During the audio feature extraction process, we employed the wav2vec 2.0 model (Baeovski et al., 2020), whose detailed architecture is shown in Fig. 2. This model performs deep analysis of raw audio signals to obtain high-resolution frame-level acoustic features. All audio segments were normalized, and the sampling rate was uniformly set to 16,000 Hz. The audio waveforms were then divided into overlapping short frame units, allowing the extraction of fine-grained acoustic features such as pitch variation, intonation changes, and rhythmic patterns, which serve as important indicators for identifying emotional states and depressive tendencies.

Given the substantial variability in the lengths of audio samples, a median-based normalization strategy was designed. Specifically, the length of each audio representation was adjusted to match the median number of frames across the dataset. This approach preserves the essential information within the audio sequences while significantly reducing the need for truncation and zero-padding, thereby maintaining the integrity of the original signals to the greatest extent.

4.3.2. ALBERT-large

In the domain of text modality analysis, this study employs the lightweight pretrained language model ALBERT-large (Lan et al., 2019) for deep semantic modeling, with its detailed architecture illustrated in Fig. 2. Leveraging parameter sharing and embedding matrix factorization, the model substantially reduces the number of parameters (only 12M) while maintaining semantic understanding accuracy. This design greatly enhances computational efficiency and training stability, making it particularly well-suited for sentiment analysis tasks on small- and medium-scale corpora. In the implementation process, text samples were input into the ALBERT model to extract hidden semantic vectors containing contextual information, which were used as the primary textual features.

4.3.3. Parameter settings

For model parameter settings, Table 5 provides the detailed configurations of each module within CEMTNet, and Table 6 summarizes the hyperparameters used during the training process of CEMTNet. The model was optimized using the Adam optimizer, with the initial learning rate set to 2×10^{-5} , and the CosineAnnealing strategy

Table 5
Parameter settings of the CEMTNet model.

Parameter	Value
Audio encoder	wav2vec 2.0
Text encoder	ALBERT-large
CCIM MLP hidden size	256
CCIM MLP layers	2
EMSA prompt vectors	Learnable, dim 256
EMSA attention type	Scaled dot-product, positional bias, attention mask
CE-TM residual fusion	Yes
CE-TM LayerNorm	Yes
CE-TM dropout	0.3

Table 6
Training hyperparameters for CEMTNet.

Parameter	Value
Optimizer	Adam
Initial learning rate	2×10^{-5}
Learning rate scheduler	CosineAnnealing
Loss function	Binary cross-entropy
Batch size	16
Number of epochs	85
GPU	NVIDIA RTX 4090D
Sampler	WeightedRandomSampler

(Loshchilov & Hutter, 2016) was adopted to dynamically adjust the learning rate and stabilize the training process. The loss function was binary cross-entropy. Training was conducted on an NVIDIA RTX 4090 GPU (24GB memory), with a maximum batch size of 16 and a total of 85 training epochs. In addition, WeightedRandomSampler (Chen et al., 2024b; Gui et al., 2024; Haixiang et al., 2017) was applied during training to dynamically balance the sample distribution between each batch and the imbalanced class distribution, and the model weights corresponding to the highest validation F1-score were saved.

4.4. Baselines

To evaluate the effectiveness of our method, we compared the reported results with recent approaches on depression detection tasks using these two datasets. The compared models are as follows:

EF-LSTM: Early Fusion LSTM concatenates multimodal inputs and uses a single LSTM to learn contextual information.

MuT (Tsai et al., 2019): Employs directed cross-modal attention to model interactions between different modalities and time steps, enabling adaptive information flow across sequences.

CMMLD (Lim et al., 2025): A lightweight multimodal depression detection model using MLP-Mixer (audio), Transformer (text), and cross-attention fusion, optimized for cross-lingual robustness and efficiency.

MAFIM + CAMFM (Xue et al., 2024): Integrates multi-level audio features (LLDs, mel-spectrogram, wav2vec) and BERT-based text embeddings via channel-attention fusion, achieving strong results on DAIC-WOZ and EATD-Corpus.

ABAFnet (Xu et al., 2024): A state-of-the-art speech-based depression detection network, fusing multiple acoustic features with type-adaptive CNN, LSTM-attention, and dynamic weighting, outperforming prior audio baselines.

DECEN (Yan et al., 2025): Incorporates depressive emotion recognition and emotion-context enhanced representation for social media depression detection, demonstrating superior performance and cross-lingual generalization.

IIFDD (Chen et al., 2024a): A multi-modal framework integrating intra-modal and inter-modal fusion for text, audio, and vision, with attention-based integration, showing **state-of-the-art** results on Chinese multi-modal datasets.

Text-guided Multimodal (CMFD + BCA) (Chen et al., 2025): One of the strongest baselines, combining cross-modal feature reconstruction,

decomposition, and bi-directional cross-attention for audio-text integration, achieving **state-of-the-art** results on DAIC-WoZ and E-DAIC.

4.5. Experimental results

To comprehensively evaluate the effectiveness of our proposed CEMTNet model, we compared its performance with a series of strong baseline methods on both the DAIC-WOZ and EATD-Corpus datasets. All experiments were conducted using 10-fold cross-validation to ensure statistical reliability and robustness under small-sample conditions.

4.5.1. Overall performance comparison

Table 7 presents the depression classification performance of various methods on the DAIC-WOZ dataset. The results show that our proposed CEMTNet achieves a significant performance improvement over all baseline methods. Specifically, CEMTNet obtains the best F1-score of 0.92, with a recall of 0.93 and a precision of 0.93. Compared with representative multimodal baselines such as EF-LSTM, MuT (Tsai et al., 2019), CMMLD (Lim et al., 2025), MAFIM + CAMFM (Xue et al., 2024), ABAFnet (Xu et al., 2024), DECEN (Yan et al., 2025), IIFDD (Chen et al., 2024a) and the state-of-the-art CMFD + BCA (Chen et al., 2025), our model consistently delivers superior results. This demonstrates the effectiveness of our cognitive consistency-guided regulation and emotion-aware structured attention, together with contrastive emotion trend modeling, in capturing semantic and emotional cues from both audio and text modalities.

Table 8 summarizes the results on the EATD-Corpus dataset. Again, CEMTNet outperforms all competing approaches, achieving the highest F1-score, recall, and precision among all compared methods. These findings further validate the generalizability and robustness of CEMTNet across different datasets and evaluation settings. By effectively modeling both semantic and affective information, our approach provides more accurate and stable depression detection under diverse real-world conditions.

4.5.2. Statistical analysis of model performance

Fig. 3 presents the training and validation accuracy curves for fold 10 in the 10-fold cross-validation. As shown, the training accuracy increases steadily and eventually converges toward 1.0, indicating that the model has effectively captured the underlying patterns of the training data. The validation accuracy shows minor fluctuations during the training process but stabilizes at a high level, reflecting the model's strong generalization ability on unseen data. Notably, as training progresses, the gap between training and validation accuracy narrows, and the two curves closely align. This suggests minimal overfitting and indicates that the model achieves an optimal balance between training performance and generalization, demonstrating excellent robustness.

In addition to the accuracy curves, Figs. 4 and 5 show the confusion matrices for fold 1 and fold 10, respectively. The confusion matrix provides detailed insight into the model's classification performance by showing the true positives, false positives, true negatives, and false negatives. For Fold 1 (see Fig. 4), the confusion matrix demonstrates a strong performance, with a high number of true positives (depressed samples correctly identified as depressed) and true negatives (non-depressed samples correctly identified as non-depressed). The false positive and false negative rates are low, which suggests that the model has a low tendency to misclassify depression states. Similarly, in Fold 10 (Fig. 5), the confusion matrix further confirms the model's robustness, as evidenced by a high number of correct classifications across both classes.

We evaluated the performance of the proposed CEMTNet model using stratified 10-fold cross-validation on the DAIC-WOZ dataset. The detailed results are presented in Figs. 6 and 7. Specifically, Fig. 6 provides a heatmap overview of fold-wise accuracy, precision, recall, and F1-score, while Fig. 7 presents the corresponding distribution of precision, recall, and F1-score across all folds as grouped bar

Table 7

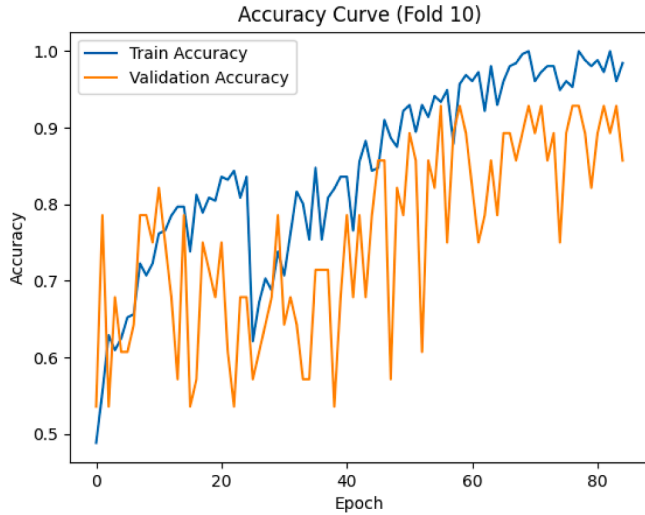
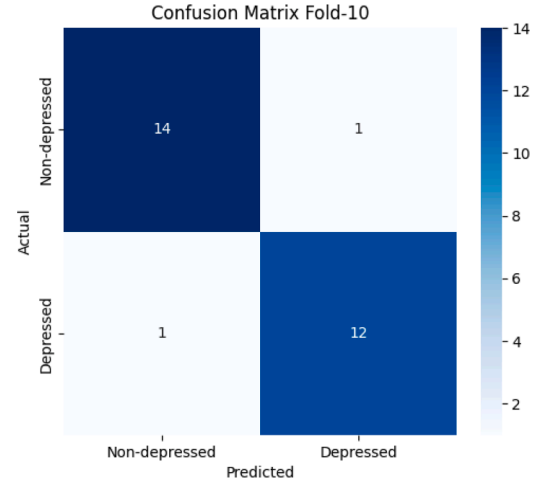
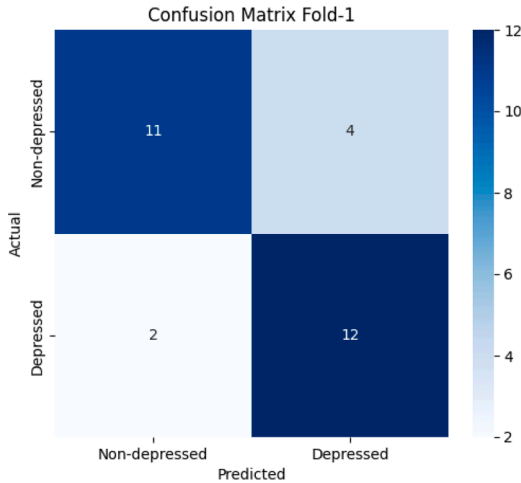
Performance comparisons on the DAIC-WOZ dataset.

Baselines	Evaluation	Accuracy $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1-score $\uparrow$
EF-LSTM	10-fold	0.67	0.68	0.70	0.67
MuT (Tsai et al., 2019)	10-fold	0.53	0.51	0.61	0.56
CMMLD (Lim et al., 2025)	10-fold	0.67	0.67	0.70	0.67
MAFIM + CAMFM (Xue et al., 2024)	10-fold	0.89	0.85	0.83	0.86
ABAFnet (Xu et al., 2024)	10-fold	0.73	0.73	0.81	0.76
DECEN (Yan et al., 2025)	10-fold	0.86	0.84	0.86	0.87
IIFDD (Chen et al., 2024a)	10-fold	0.87	0.85	0.87	0.85
CMFD + BCA (Chen et al., 2025)	10-fold	0.89	0.87	0.93	0.90
CEMTNet (Ours)	10-fold	0.92	0.93	0.93	0.92

Table 8

Performance comparisons on the EATD-Corpus dataset.

Baselines	Evaluation	Accuracy $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1-score $\uparrow$
EF-LSTM	10-fold	0.44	0.56	0.49	0.49
MuT (Tsai et al., 2019)	10-fold	0.72	0.70	0.71	0.72
CMMLD (Lim et al., 2025)	10-fold	0.81	0.82	0.81	0.82
MAFIM + CAMFM (Xue et al., 2024)	10-fold	0.72	0.73	0.75	0.77
ABAFnet (Xu et al., 2024)	10-fold	0.77	0.76	0.79	0.77
DECEN (Yan et al., 2025)	10-fold	0.88	0.84	0.87	0.87
IIFDD (Chen et al., 2024a)	10-fold	0.82	0.81	0.82	0.86
CMFD + BCA (Chen et al., 2025)	10-fold	0.87	0.88	0.87	0.87
CEMTNet (Ours)	10-fold	0.91	0.90	0.91	0.92

**Fig. 3.** Training and validation accuracy curves on the 10th fold of 10-fold cross-validation.**Fig. 5.** Confusion matrix for Fold-10.**Fig. 4.** Confusion matrix for Fold-1.

plots. CEMTNet demonstrates consistently robust classification performance across all evaluation folds. The model achieves an average precision of 85.2% ($\pm 13.5\%$), recall of 79.4% ($\pm 13.2\%$), and F1-score of 81.3% ($\pm 10.1\%$). The 95% confidence intervals for precision ([75.6%, 94.9%]), recall ([70.0%, 88.8%]), and F1-score ([74.0%, 88.5%]) indicate that the estimation is both reliable and stable. Among all folds, the best performance appears in Fold 9, with a recall of 100% and an F1-score of 93.8%. In contrast, the lowest metrics are observed in Fold 5, where the precision, recall, and F1-score are 53.8%, 58.3%, and 56.0%, respectively. This discrepancy is likely attributable to class imbalance or more challenging data distributions within this fold. Importantly, the standard deviations across all metrics remain moderate, demonstrating the strong generalization capability and robustness of CEMTNet in the face of data variability. These comprehensive results—supported by both the heatmap and bar plot—confirm that CEMTNet delivers not only high average performance but also excellent stability and reliability across stratified cross-validation folds, highlighting its practical value for robust depression detection.

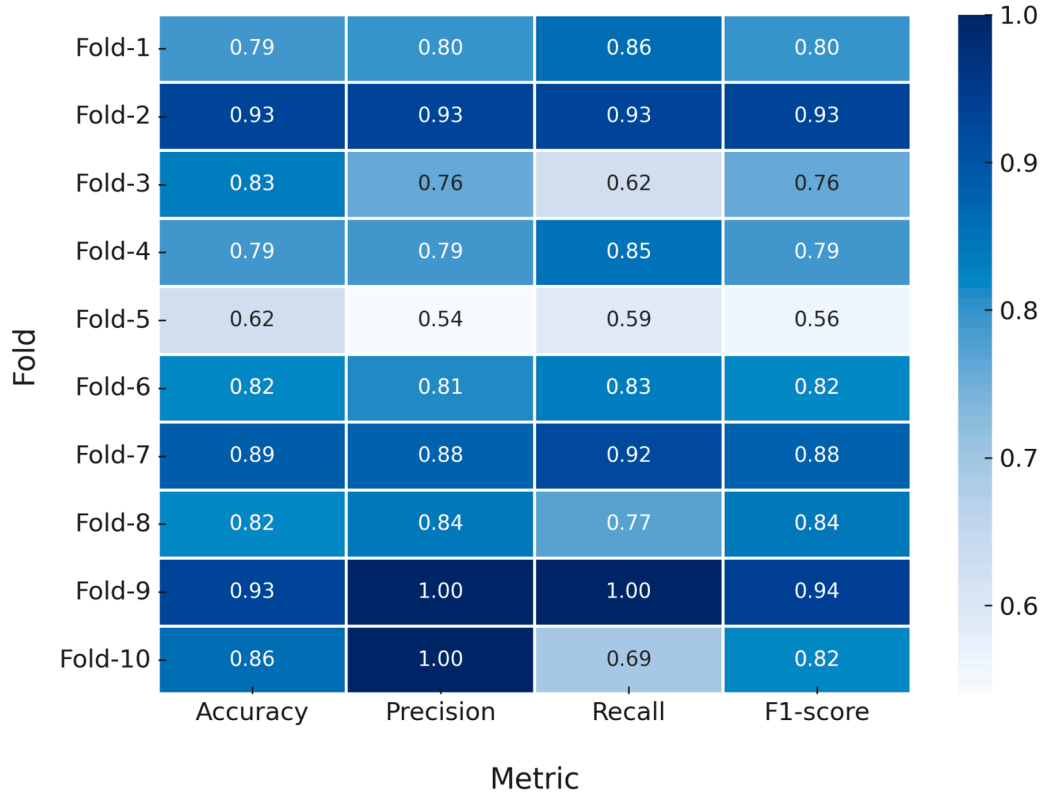


Fig. 6. Heatmap of performance metrics across 10-fold cross-validation.

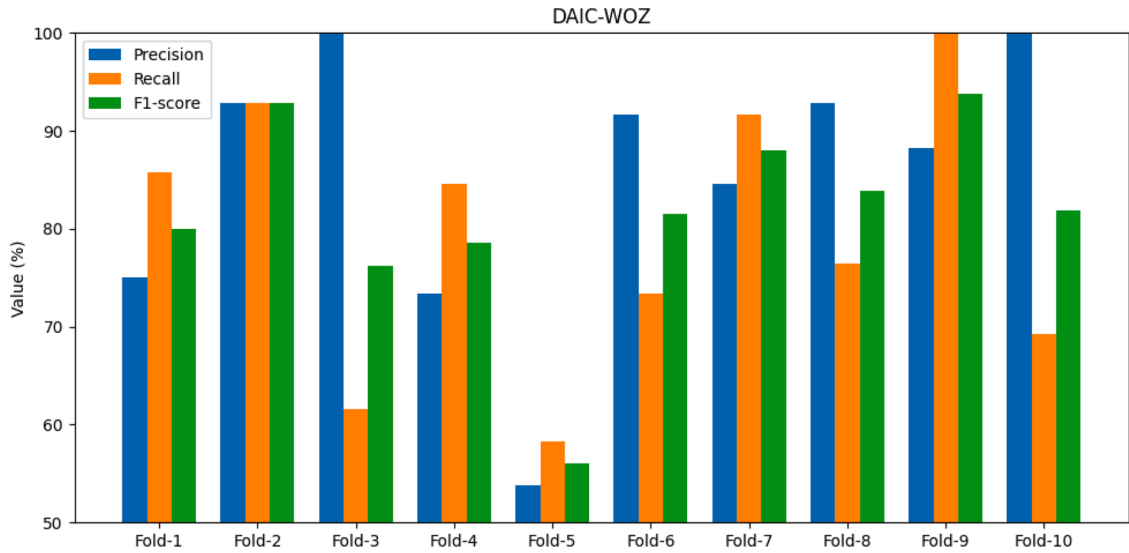


Fig. 7. Bar plot of performance metrics across 10-fold cross-validation.

4.6. Ablation study

To verify the effectiveness of the proposed modules, a comprehensive ablation study was conducted on the DAIC-WOZ dataset, as described below. The ablation results are presented in Table 9.

Effect of the CCIM: The CCIM is designed to dynamically regulate cross-modal feature fusion. To evaluate the effectiveness of the CCIM module, we compared the model's performance with and without the CCIM module. As shown in Table 8, the F1-score of "w/o CCIM" is 0.857, the recall is 0.842, and the precision is 0.873, which are all substantially

lower than those of our proposed method. This demonstrates the importance of the CCIM module.

Effect of the EMSA Module: The EMSA module introduces emotional prior information to guide the attention mechanism toward ambiguous and low-intensity emotional cues within the multimodal inputs, thereby enhancing the model's ability to recognize weak emotional signals. To assess the effectiveness of the EMSA module, we compared the model's performance with and without the EMSA module. As shown in Table 8, the F1-score of "w/o EMSA" is 0.878, the recall is 0.878, and the precision is 0.878. These results are lower than

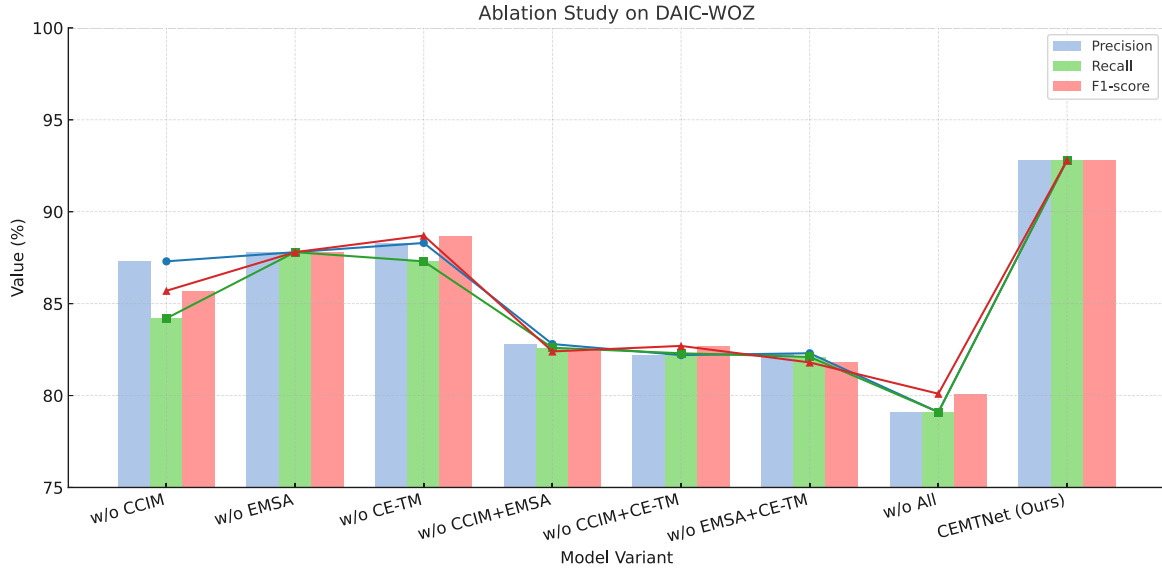


Fig. 8. Visualization of module contribution in the ablation study on DAIC-WOZ.

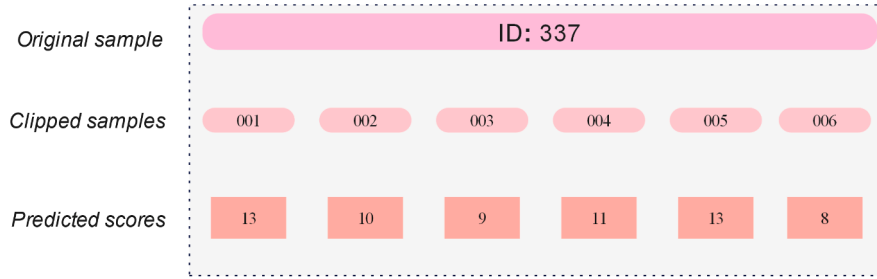


Fig. 9. Illustration of the case study. The ground truth score for the original sample is 10, while the average predicted score across all segmented fragment-level samples is 10.67. This close alignment between the predicted and actual scores demonstrates the model's ability to capture fine-grained emotional variations across temporal segments, highlighting its potential for accurate depression severity estimation in real-world clinical scenarios.

Table 9
Ablation study on the DAIC-WOZ dataset.

Method	Precision ↑	Recall ↑	F1-score ↑
w/o CCIM	0.873	0.842	0.857
w/o EMSA	0.878	0.878	0.878
w/o CE-TM	0.883	0.873	0.887
w/o CCIM + EMSA	0.828	0.826	0.824
w/o CCIM + CE-TM	0.822	0.823	0.827
w/o EMSA + CE-TM	0.823	0.821	0.818
w/o CCIM + EMSA + CE-TM	0.794	0.791	0.801
CEMTNet (Ours)	0.928	0.928	0.928

those of our proposed method, confirming the significance of the EMSA module.

Effect of the CE-TM Module: The CE-TM module aims to capture the temporal emotional dynamics in the multimodal interaction process through cross-time-step contrastive learning, thereby improving the model's ability to perceive emotional fluctuations and turning points. To evaluate the effectiveness of the CE-TM module, we compared the model's performance with and without the CE-TM module. As shown in Table 8, the F1-score of "w/o CE-TM" is 0.887, the recall is 0.873, and the precision is 0.883. These results are lower than those of our proposed method, demonstrating the effectiveness of the CE-TM module.

Furthermore, we analyze the combined effects of module interactions. The comprehensive ablation results are visualized in Fig. 8, which intuitively illustrates the performance variations under different

module configurations. The performance further decreases when multiple modules are simultaneously removed, such as CCIM + EMSA ("w/o CCIM + EMSA", F1-score: 0.824), CCIM + CE-TM ("w/o CCIM + CE-TM", F1-score: 0.827), and EMSA + CE-TM ("w/o EMSA + CE-TM", F1-score: 0.818). The most significant performance drop occurs when all three modules are removed ("w/o CCIM + EMSA + CE-TM"), resulting in an F1-score of 0.801, recall of 0.791, and precision of 0.794. These results further confirm the complementary effects and necessity of jointly incorporating emotional consistency reasoning, weak emotional cue enhancement, and temporal trend modeling to achieve robust multimodal depression detection. Beyond demonstrating these complementary effects, it is noteworthy that removing individual modules or their pairs results in comparable performance drops. This uniform decline stems from the highly coupled, sequential nature of CEMTNet. CCIM establishes foundational emotional alignment, enabling EMSA to extract fine-grained cues, which CE-TM subsequently uses for temporal modeling. Operating as a synergistic chain, ablating any single module breaks the continuous feature refinement pipeline, causing the network to revert to a basic performance floor. Thus, all three components are strictly indispensable.

4.7. Case study

To further validate the capability of the proposed model in addressing emotional inconsistency, semantic misalignment, and dynamic emotion modeling, we conducted an in-depth depression case analysis based

on the DAIC-WOZ dataset. The detailed experimental results are presented in Fig. 9. A representative sample with a depression score of 10 (Sample ID: 337) was selected to accurately assess the model's ability to predict overall depression severity. The sample was segmented into multiple sub-samples at the fragment level, and predictions were made on each fragment individually using the proposed method.

The results show that the predicted scores of most sub-samples fluctuate around 10, with an average predicted score of 10.67. This value closely matches the ground truth score, fully demonstrating the effectiveness and reliability of CEMTNet in depression assessment. In contrast, other methods produced predictions that deviated significantly from the true score of 10, further validating the practical value of the proposed approach.

5. Discussion

The comparative experiments presented above demonstrate the significant effectiveness and advantages of the proposed model. To further clarify the contribution of each innovative module, a series of ablation studies were conducted. The results from these ablation experiments clearly validate the effectiveness of the designed modules. In addition, comprehensive experiments on both the DAIC-WOZ and EATD-Corpus datasets show that the proposed model achieves an average recall of approximately 0.9. The high recall indicates that the model is designed to maximize the identification of potential individuals with depression. However, the relatively lower precision suggests that a certain proportion of non-depressed subjects are misclassified as depressed.

Although the model exhibits a certain level of misclassification, the advantage of high recall lies in its capacity to identify a greater number of potential depression cases at an early stage, thereby providing initial alerts and supporting further clinical assessment. Despite achieving the best performance on the EATD-Corpus dataset, there remains considerable room for improvement. We speculate that this may primarily stem from class imbalance within the dataset. Although the WeightedRandomSampler technique was applied during training to alleviate class imbalance, the results still suggest that the subtle emotional features of depression may be easily overshadowed by the dominant patterns of the majority class. Moreover, depressive symptoms often manifest as subtle changes in both verbal and non-verbal cues, such as slight variations in speech rate, tone, or specific word choices. To better capture these nuanced emotional signals, the CCIM and EMSA modules were specifically designed to address cross-modal emotional inconsistency and semantic misalignment. The experimental results confirm the effectiveness of these modules. However, when processing noisy or incomplete contextual data, the model's performance may noticeably decline. Such situations call for more refined and robust feature extraction techniques to enhance sensitivity to subtle emotional cues.

Additionally, a notable decline in model performance was observed in Fold 5 of the cross-validation experiments. Upon analysis, we believe this issue may be attributed to data quality instability, such as background noise in the audio data or incomplete text transcriptions. These findings underscore the importance of robust data preprocessing techniques such as denoising, normalization, and text alignment for ensuring model stability and reliability.

Another critical challenge is the risk of misdiagnosis, particularly false negatives. Given the potentially serious consequences of false negatives in real-world applications, we recommend adopting comprehensive enhancement strategies, such as introducing confidence-based dynamic thresholds, to improve the model's reliability and safety in practical settings. While CEMTNet demonstrates strong performance, deploying it as a reliable clinical tool still faces practical limitations. The current framework cannot explicitly untangle depression from frequent psychiatric comorbidities such as anxiety. Moreover, its accuracy remains susceptible to domain shifts when tested on new datasets collected under different acoustic environments. Future research dedicated to domain-invariant learning and fine-grained symptom disentanglement will be

essential for real-world adoption. Furthermore, depression detection research involves processing highly sensitive personal information, including audio and textual data. Therefore, ethical considerations are paramount throughout data collection and research implementation. Informed consent must be obtained from participants in advance, and data processing must comply with ethical guidelines to protect participants' privacy and data security.

6. Conclusion

In this study, we propose CEMTNet, a novel multimodal cognitive-emotion consistency framework designed to address key challenges in depression detection, including cross-modal emotional inconsistency, subtle emotional cue extraction, and dynamic emotional variation modeling. The architecture incorporates three innovative components: CCIM dynamically regulates the fusion weights between speech and text modalities through a consistency-guided mechanism to mitigate cross-modal emotional inconsistency; EMSA introduces an emotion-aware structured attention mechanism to precisely capture subtle and fine-grained emotional cues in both speech and text modalities; and CEMTNet explicitly models the temporal dynamics of emotional expression through a contrastive learning framework, enhancing the model's capability to track depressive emotional trends. Experimental validation on the DAIC-WOZ and EATD-Corpus datasets demonstrates that CEMTNet significantly outperforms state-of-the-art baseline models, achieving notable improvements in depression recognition accuracy. The ablation study further confirms the indispensable contributions of each module and highlights how their synergistic integration enhances representational discrimination and emotional coherence modeling.

Despite its robust performance, the model's effectiveness remains dependent on the quality of input data, such as background noise in audio signals and incomplete text transcriptions. This underscores the need to develop advanced preprocessing pipelines to ensure consistent and high-fidelity multimodal data inputs. From an ethical perspective, given the sensitivity of personal data involved in depression detection tasks, this study emphasizes the critical importance of implementing strict privacy protection protocols and obtaining informed consent. In future research, several directions will be further explored. First, the integration of physiological signals (e.g., heart rate variability, electrodermal activity) and behavioral signals (e.g., facial expressions, motor behaviors) will be considered to enrich the dimensionality of emotional state modeling and enhance the system's capacity to capture complex emotional dynamics. Second, leveraging the strengths of large language models (LLMs) in cross-modal semantic alignment and contextual understanding, we will investigate their potential for knowledge transfer and few-shot adaptation in depression detection, particularly for recognizing subtle and implicit emotional cues (Hu et al., 2024; Lu et al., 2025). In addition, prompt engineering shows promising task transferability and adaptability in mental health applications, which may improve the model's flexibility and clinical applicability across diverse scenarios. Furthermore, improving cross-lingual and cross-cultural generalization remains an important challenge, contributing to the development of more universally applicable depression recognition systems.

CRedit authorship contribution statement

Yujie Huo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology; **Hongyu Gao:** Visualization, Methodology, Resources, Writing – review & editing; **Weng Howe Chan:** Supervision, Resources; **Ahmad Najmi Bin Amerhaider Nuar:** Supervision.

Data Availability

Source code and data are available at <https://github.com/ghy-yhg/CEMTNet>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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